

JARVIS

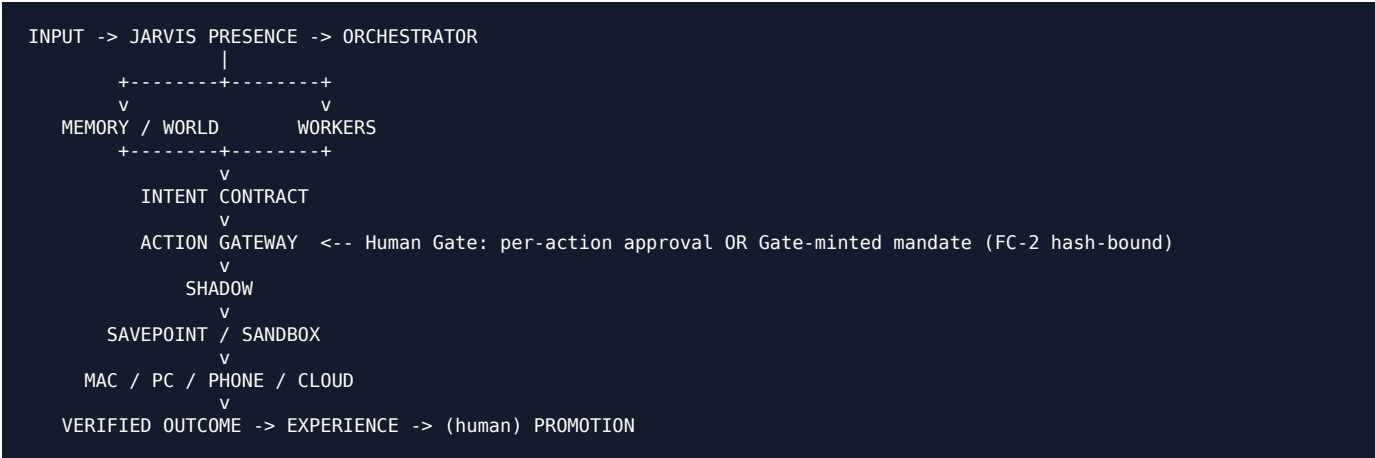
Master Architecture & Delivery Roadmap v5.1

EXPANSION ERA REBASELINE, RECONCILED TO THE REPOSITORY - 01 SEPTEMBER 2026

CORE POSITION

JARVIS is a local-first personal intelligence environment: one persistent identity, a fast conversational presence, governed specialist workers, durable execution across trusted nodes, a spatial world model, and an action boundary that verifies consequences before meaningful side effects. **The Human Gate is the only mutation path.** Every v5 addition strengthens that boundary or it does not ship.

What v5 is: v4 with its facts corrected against the repository (Phases 22-24 are frozen, not future; Phase 25 Productionization exists and was missing), its numbering de-collided, its authority model tightened at three holes, and its delivery law extended so that DONE cannot be claimed over synthetic data, lab branches, or open residuals.



Prepared for	Claude Code / Codex execution, after Prince's verdict. Architect of record: Fable.
Supersedes	v5.0 (same day; three confirmations were open) and JARVIS Master Architecture & Delivery Roadmap v4.0 (01 Sep 2026) in full; v3/v3.1 for post-Phase-24 planning only.
Repository truth	Frozen Phases 1-24. Last recorded working session ~08 July 2026 (Phase 24 frozen, WorkflowBox v1a-v1c complete, E-011 prompt written). Test baseline at that session: ~5,900 passing. Nothing in this document reopens a frozen phase.
Status	Private working architecture / execution brief. Read section 0 first; every change from v4 is listed there with its reason. v5.1 closes the three confirmations: primary machine M1 Max 32GB; move complete, Phase 25 gate OPEN; Playroom confirmed and seam stated.

DONE MEANS DONE. NO PHASE CLOSES ON SCAFFOLDING, ON A LAB BRANCH, OR OVER SYNTHETIC DATA.

0. v4 -> v5 Change Register (read this first)

Every tweak from v4 is listed with the reason. v4 was written from the 03 June 2026 Expansion Era refresh and the EXPANSION_ERA_V2 sequencing. It did not see the June/July build sessions. Three classes of correction follow: **facts** (what the repo actually contains), **Gate-tightening** (three authority holes), and **DONE-discipline** (gates v4 stated but could not enforce).

#	v4 said	v5 says	Why
1	Baseline: Phase 21 is the latest frozen phase; 22-26 are future work.	Phases 1-24 are frozen. 22 = Voice Authority (T0-T3, standing consent), 23 = Vision + Sensing Realization (I1-I5), 24 = MCP Gateway (FC/GATE/EoP, zero residuals).	Fact. v4's Phase 22E ('implement standing-consent store, T2 UI confirm, T3 hard block') is already built and frozen. Sent to CC as-is, v4 reopens three frozen phases by accident.
2	New phases numbered 22-26.	New phases numbered 26-30 (Presence, Spatial, Local Intelligence, Room, Node Fabric), provisional until audit step 0 confirms no collision. Tracks A-G stay unnumbered.	Collision. v4's own rule ('do not renumber history') is violated by v4's own numbering. Phase 25 is already named in the registry/recap as Productionization.
3	No productionization phase anywhere in the sequence.	Phase 25 - Productionization is the first numbered phase after the runway: synthetic-to-real audit, realize-with-drill, voice stack live, target-hardware bootstrap, 23F camera proof, UI validation on real data, packaging, 7-day soak.	DONE means DONE. v4 stacks five new capability phases on a system that has never run end-to-end on real data. That is scaffold multiplication. Every later closeout gate ('real integration', 'target hardware') needs the Phase 25 baseline to be checkable at all.
4	Program U starts immediately after the audit.	Pre-Phase-25 runway inserted: audit -> stabilize -> E-011 voice consolidation -> real-vs-synthetic check -> Program U.	The July runway was sequenced 'do not reorder': the capstone renders voice state, and two duplicated TTS chains render incoherently. E-011 has a prompt written (e011-voice-consolidation.md).
5	Program U closes 'operationally validated' at U.8 with real API/event data.	Program U closes at Execution Enabled . U.8 moves into Phase 25 as slice 25F.	Real data does not exist until Phase 25. A UI program cannot self-certify over synthetic feeds; v4 gate 1 forbids exactly that.
6	22E 'Voice authority T0-T3': implement standing-consent store, revocation, T2 confirm, T3 block.	Deleted. Cited as frozen Phase 22.	Already built (voice-operating-mode/, config/voice/standing-consent.yaml).
7	22C local wake word lives in the Presence phase.	Wake word moves to Phase 25 slice 25C. The multi-turn conversation state machine stays in Presence (26C).	openWakeWord is already wired and not listening; making it listen is productionization, not new capability.
8	22G voice-node protocol in the Presence phase.	Moves to Phase 30 (Node Fabric) as 30F.	Its exit criterion ('same utterance, same authority decision regardless of inlet') needs a second inlet, i.e. the phone node.
9	'Governed MCP - Adopt: MCP is capability transport.'	Split: MCP server side is frozen Phase 24 (stdio only). MCP consumer side (JARVIS calling external MCP servers) is Track B slice B2b with its own threat model in the Phase 24 method. HTTP/SSE transport is a separate, heavier threat model.	Phase 24 section 7 declares consuming external MCP and network transport out of scope. Conflating the two invites reopening 24.
10	Hardware: '32GB M1 Max' as the Mac; benchmark on 'actual 32GB M1 Max'.	Confirmed 01 Sep 2026: M1 Max, 32GB unified memory is the primary machine (v4 was right; the July M3 Pro decision was superseded at purchase). The July 18GB envelope (8B workhorse / 14B ceiling) is therefore stale. Prior for 28A, unmeasured: 8B workhorse; 14B comfortable; 27-32B-class feasible at 4-bit; 70B still out. 28A records the truth.	Was a conflict between v4 and the last recorded decision; resolved by Prince. The envelope is a prior, not a claim, until benchmarked.

#	v4 said	v5 says	Why
1 1	T2 requires UI confirmation 'unless future policy explicitly narrows and re-ratifies class'.	'Unless' deleted. T2 always requires UI confirmation. Moving an action class from T2 to T1 is itself a T3 governance change: manual, registry entry, closeout test.	Gate-tightening. An open-ended reclassification clause is a hole a model could drive policy through.
1 2	Workers 'may perform bounded actions inside explicit mandates' - who mints a mandate is unstated.	Mandates are Gate artifacts: MandateGrant v1 , minted only through the frozen approval lifecycle (a mandate proposal is a T2 UI-confirm), bound to actor + action class + target class + budget + expiry, revocable, receipted per execution. Workers cannot mint, widen or renew one. Missing/expired/ambiguous = deny.	Gate-tightening. Same shape as Phase 22 standing consent and Phase 24 FC-2: the Gate moves from per-action to per-mandate, it does not move out of the human's hands.
1 3	'Persistent scheduler outside model' wakes workers.	Kept, as an additive registry amendment wrapping frozen Phase 8: the scheduler may <i>wake</i> a worker; it may never approve or execute. Whatever the woken worker does is T0 read, T1-under-mandate, or T2 propose.	Phase 8 froze 'scheduler is suggestion-only'. Wake is not authorize; the amendment must say so in a test, the way FC-2 wrapped Phase 18.
1 4	Memory/procedure PromotionDecision may PROMOTE automatically (reviewer field can be a model).	Automation may quarantine, score, replay, canary and propose . A human promotes, through the Gate, via a batched promotion inbox. Promoted procedures are T2 knowledge.	Frozen 'no silent writes' (Phases 5, 21). A promoted procedure is executable knowledge; letting a model promote it is the Sleepwalk failure the dissertation is about.
1 5	Section 21 'superseded decisions' lists old positions as replaced.	Each becomes a registry amendment naming the frozen invariant it touches and the additive/wrap shape required.	'Superseded' is not a roadmap verb. Frozen invariants change only through the Enhancement Registry with a closeout test.
1 6	Delivery Law: 12 gates.	17 gates. Added: on main; closeout doc + registry flip; hooks run in full / no --no-verify; runtime.runTool = 2 canary at every closeout; residual rule (a phase freezes only with zero residuals, or a named hardware-gated residual, or a residual scheduled as the immediate next slice).	These are the repo's actual conventions since Phase 21. v4 stated the philosophy; these are the checks CC can run.
1 7	Status vocabulary stated without mapping.	Mapped to registry states (PROPOSED / IN-PROGRESS / COMPLETE / DEFERRED / PENDING-HARDWARE) so CC never invents a parallel vocabulary.	One source of truth for status is the point of the registry.
1 8	Frozen list covers Phases 1-20 style invariants only.	Extended with Phase 21-24 invariants, the registry-as-only-entry rule, and the single-agent rule for boundary work.	Fact. The most valuable invariants in the repo (I1-I5, FC-1/2/3, GATE-1..5, the import allowlist) were absent.
1 9	External references cited without dates/status.	Verified 01 Sep 2026: Hermes v0.21.0 released 31 Aug 2026; AgentCore runtime instances GA Aug 2026 (14-day sessions); OpenBot first tag v0.0.1 on 17 Aug 2026, self-described alpha. Register now marks 'pattern-adopt only' where the runtime is alpha or high-churn.	Search-before-claim rule. A P0 pattern from an alpha runtime is fine; a P0 dependency on it is not.
2 0	'The system sat untouched for roughly three months.'	Roughly eight weeks (last session ~08 July 2026).	Accuracy; it changes how much drift the audit should expect.
2 1	'JARVIS is single-principal. Playroom is the multi-principal product.'	Confirmed 01 Sep 2026. Playroom is Prince's multi-principal, multi-provider room product (Master Roadmap v1.0, 24 Jul 2026). Repository state: a landing page on a vinext starter, no fabric code, P0 never opened. Section 3 now states the JARVIS/Playroom seam; the Playroom build decision belongs to the Clearance roadmap, not this document.	Name now known. JARVIS must not import Playroom's roster/permit trust model, and this roadmap must not widen into a fourth product.
2 2	Milestone demos.	Added: Productionization soak demo, Mandate revocation demo, Numbering-audit proof.	Each new discipline in v5 needs a demo that proves it, or it is a sentence.

#	v4 said	v5 says	Why
2 3	Instructions to Fable.	Rewritten as instructions to CC/Codex: the numbering-collision check, the exact files to read, the count check, the hardware check.	Fable is the author now; CC is the reader.

THE THREE CONFIRMATIONS - RESOLVED 01 SEP 2026 (v5.1)

- 1. **Primary machine: M1 Max, 32GB unified memory.** Recorded in section 1; 25D bootstraps on it; 28A benchmarks it. The July 18GB envelope is retired.
- 2. **The move happened.** The Phase 25 gate is OPEN. Phase 25 follows the runway and Program U with no further external gate.
- 3. **Playroom is real** (roadmap v1.0, 24 Jul 2026; code = landing page). The seam is stated in section 3. No CC slice is blocked on it.

1. Executive Rebaseline

WHY v5 EXISTS

The May/June architecture was correct on local-first routing, governance, adapter conformance, telemetry and explicit state. Between 03 June and 08 July the repository then moved further than any planning document recorded: the voice authority tiers shipped (Phase 22), a real video/vision pipeline with five audited invariants shipped (Phase 23), and JARVIS became an MCP server whose Human Gate is provably non-bypassable from outside (Phase 24, twelve-path drill, zero residuals). WorkflowBox shipped in three slices. v4 was drafted from the June documents and re-planned that same ground under the same numbers. v5 keeps v4's target architecture and adoption research, which are sound, and rebases them on the repository as it is.

BASELINE TRUTH (repository, last session ~08 July 2026)

Frozen: Phases 1-24. 1-20 governance and operationalization substrate (31 May). 21 Expansion workflows 21A-21K plus the frozen cockpit surfaces and pipeline spine (03 June). 22 Voice Authority: T0-T3 tiers, standing-consent store, PTT runtime, TTS chain; wake word wired, not listening; fallback status recorded honestly under E-010. 23 Vision + Sensing Realization 23A-23G: yt-dlp/ffmpeg/faster-whisper packet pipeline, payload-rejecting multimodal schema, invariants I1-I5, consent-gated webcam (23F camera proof PENDING-HARDWARE). 24 MCP Gateway 24A-24E: server-derived canonical effect (FC-1), hash-frozen proposal with approval-time re-validation (FC-2), token-bound identity (FC-3), GATE-1..5, EoP/ID threat table all GREEN, E-016/E-017 closed, runtime.runTool = 2 production call sites throughout.

Built, not frozen: WorkflowBox v1a-v1c (single source of truth, lane list + SVG mind-map, amber door as a type seam, not wired). Enhancement Registry E-001..E-018. UI_CAPSTONE_BRIEF.md banked.

Open: E-011 (voice surface consolidation, prompt written - THE live next slice), E-005, E-009, E-012 (real fallback beyond demo-director), 23F-CAMERA-PROOF (hardware).

Hardware (confirmed 01 Sep 2026): primary machine is a MacBook Pro M1 Max with 32GB unified memory; the Windows ASUS ROG Strix G15 (15.8GB) becomes the execution node in Phase 30. JARVIS has never been bootstrapped on the Mac - that is Phase 25D, and it is the first slice on the new machine. Until then the Unreal gauntlet and JARVIS suites still run separately (memory starvation was the root cause of the flake tail on the ASUS).

Decision	v5 position
Architecture style	Compose mature external runtimes behind stable JARVIS adapters. Do not fork or rebuild agent platforms. Every external enters behind a driver with a kill switch and conformance tests; pattern-adopt where the runtime is alpha.
Identity	JARVIS remains one singular personal presence. Specialist Workers are organs underneath, never competing personalities on the main surface.
Authority	The frozen approval lifecycle (Phase 18) plus FC-1/FC-2/FC-3 (Phase 24) remain authoritative. v5 adds fail-closed policy, consequence verification and Gate-minted mandates. No model, worker, scheduler or node can grant, widen or renew authority.
Execution	Local-first, multi-node. The M1 Max 32GB is the primary brain once bootstrapped (Phase 25D); the Windows ASUS is a GPU/render/execution node; phone is supervision and sensors; cloud is optional borrowed compute.
Learning	No unverified self-modification. Experiences become candidates, pass provenance, replay, verification and canaries, then a human promotes.
Spatial	All sensors update one canonical Spatial World Model. RuView stays observe-only until real-hardware evidence earns trust.
Delivery	Foundation work may exist as a slice and can never close a feature or phase. Operationally Validated on target hardware, on main, with a closeout doc, is the minimum meaning of DONE.
Sequencing	Runway -> Program U -> Phase 25 Productionization -> Phases 26-30 -> Tracks A-G. Nothing jumps the queue; new ideas enter the registry, not the roadmap.

2. Delivery Law: DONE Means Operationally Done

BINDING RULE

A contract, schema, fake adapter, UI mock, injected runner, or passing unit test is not a finished capability. It is foundation. A capability closes only when the real runtime exists, the real happy path and failure paths have been exercised on target hardware, all authority/privacy invariants hold, the code is on main, and the user can actually use the feature.

Gate	Required evidence before a slice/feature may close
1. Real integration	The real provider/device/runtime is installed and configured where the feature requires one. No injected fake may be the final runtime proof.
2. End-to-end execution	At least one real task travels from user/event input through routing, policy, execution, verification and visible result.
3. Failure exercise	At minimum: timeout/offline, denied permission, malformed input, cancellation/interruption, and dependency failure are demonstrated and tested.
4. Safety/authority	No bypass of approval lifecycle, standing consent, mandate, action policy, trust class or Shadow/Savepoint boundary. Fail closed when state is unknown.
5. Privacy	No forbidden raw payloads in telemetry/UI. Secret-bearing paths are tested with realistic canaries.
6. Observability	Trace/run/task IDs, model/tool/runtime selection, latency/cost, action decision, failure class and outcome are inspectable.
7. Verification	Process success and outcome success are distinct. Side effects receive verification reads or post-state checks where possible.
8. Performance	Target P50/P95 budgets measured on the actual primary machine. Regressions are bugs.
9. UX surface	If user-facing, the final Command Center/voice/mobile/spatial surface exists and is usable; no temporary developer-only control counts.
10. Recovery	Rollback/compensation/resume semantics are proven for the class of action.
11. Full regression	Unit + integration + property + E2E + security/disabled-feature + adapter-conformance suites pass, including every frozen-phase battery.
12. Closeout truth	Closeout states exactly what is enabled, disabled, unverified and deferred. No aspirational language may be labelled complete.
13. On main (new)	Code that exists only on a lab branch, in a worktree, or entangled with another checkout is not shipped. Phase 22's voice layer lived in a lab commit for a week while the recap said COMPLETE; that is the precedent this gate exists for.
14. Closeout artifacts (new)	A closeout document under docs/audits, the registry rows flipped, the recap updated, and the suggested commit message - in the same commit. No co-author trailers.
15. Hooks in full (new)	lint-staged, eslint, tsc --noEmit and vitest run in the hook and pass. --no-verify is never used; a flake is fixed or its timeout raised through a registry entry (the E-013/E-015 precedent).
16. Mutation-path canary (new)	runtime.runTool has exactly 2 production call sites (tool-approvals, tool-continuation) and resumeApproval is the sole executor - asserted at every closeout, not only in Phase 24. The six-stage pipeline spine stays byte-identical to its Phase 21 baseline.
17. Residual rule (new)	A phase may freeze only with zero residuals, or with a residual that is (a) hardware-gated and named (23F-CAMERA-PROOF), or (b) scheduled as the immediate next slice and referenced in the freeze commit (the 24E/24D-4 precedent). Unnamed or unscheduled residuals block the freeze.

STATUS VOCABULARY (mapped to the registry)

Foundation Complete -> Workflow Complete -> Realized -> Execution Enabled -> Operationally Validated.

Only Operationally Validated counts as DONE for a feature. Production-grade is a separate, higher bar and is claimed only after Phase 25H. These map onto the registry states as follows and nothing else may be used:

Registry state	Meaning	Roadmap vocabulary it may carry
PROPOSED	In the registry, no slice opened.	none
IN-PROGRESS	A slice is open on a branch/worktree.	Foundation Complete, Workflow Complete
DEFERRED	Consciously parked with a stated gate.	any, with the gate named
PENDING-HARDWARE	Code complete; proof needs a device not present.	Realized or Execution Enabled - never Operationally Validated
COMPLETE	Closeout doc written, on main, hooks green.	Realized, Execution Enabled, or Operationally Validated - the closeout must say which

3. What Remains Frozen and Must Be Preserved

Substrate invariants (Phases 1-20, unchanged from v4)

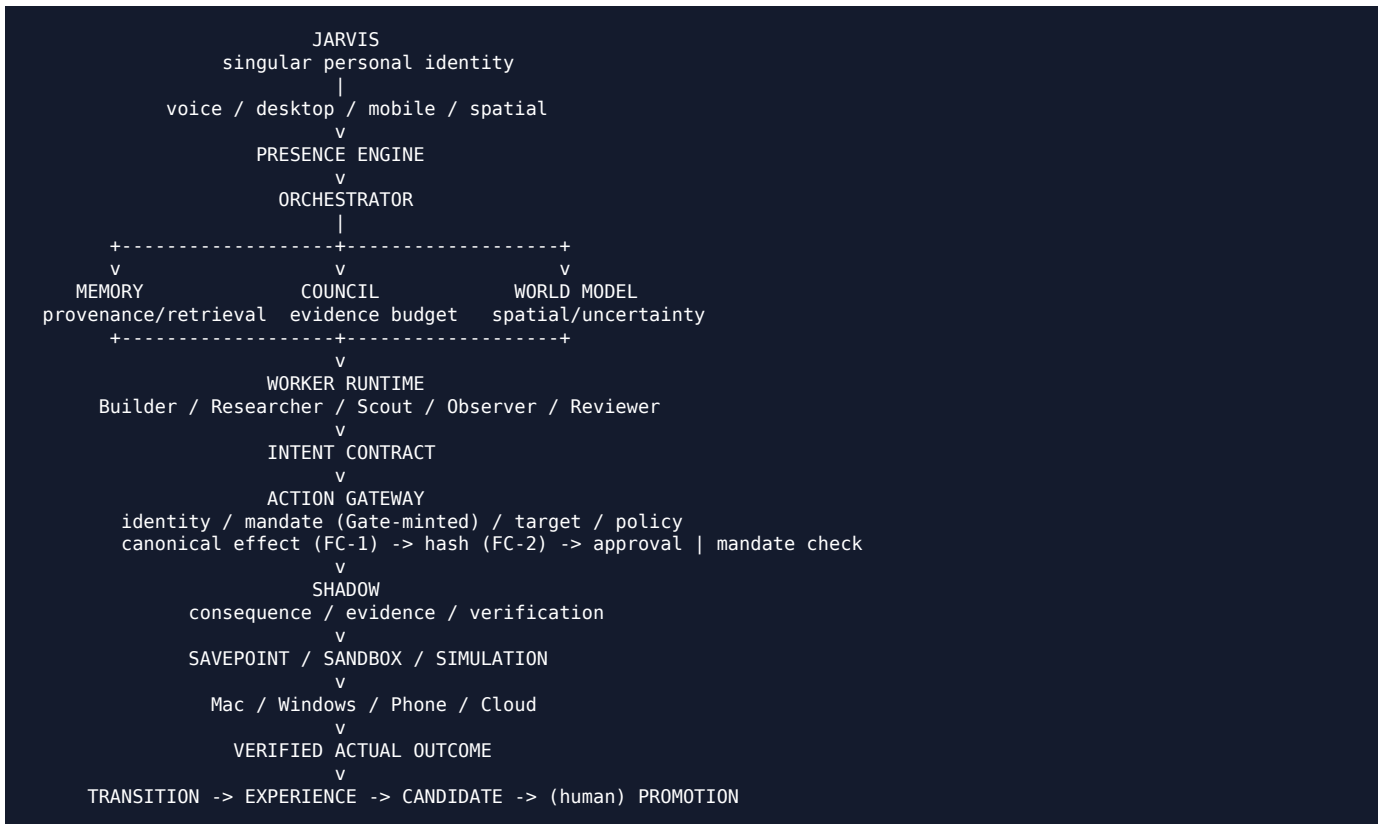
- Local-first by default; cloud is opt-in, budgeted, logged and replaceable.
- Model-agnostic provider registry and capability-based routing; models are resources, not identities.
- Deterministic tools before language models.
- One-room-first environmental scope until the room runtime is demonstrably stable.
- Adapter conformance: fake/mock and real adapter must satisfy the same contract before real hardware/providers are enabled.
- Approval lifecycle remains the single route to meaningful side effects; the v5 additions strengthen it rather than bypass it.
- No raw prompts, frames, secrets, project bodies or sensitive payloads in generic telemetry/Command Center surfaces.
- Prompt/policy/versioned configuration is code: git-tracked, diffable, reversible.
- Every significant model/tool/routing decision is observable and replayable as metadata/evidence.
- Cost caps, loop budgets and explicit stop/kill semantics remain mandatory.
- UI/graphs remain evidence-first. Graphs may explain execution but do not secretly become an execution engine.
- JARVIS is single-principal. **Playroom** (confirmed) is the multi-principal, multi-provider product; do not import its trust assumptions wholesale. **Seam**: if JARVIS ever joins a Playroom room, it joins as an external agent of a foreign fabric - the Phase 24 boundary in reverse - and T2/T3 remain UI-confirmed on JARVIS's side regardless of what the room's permit allows. Playroom's own fabric (signed permits, deny-by-default server-side evaluation, receipts on a hash chain, postage budgets) is the Clearance doctrine restated; whether it is built on the Clearance engine or beside it is a Clearance-roadmap decision, and this document does not widen to carry it.
- Scheduler is suggestion-only (Phase 8); room devices default observe_only; CAI execution blocked until explicitly opened; demo mode cannot touch live data.

Frozen by Phases 21-24 (added in v5 - cite, never reopen)

Phase	Frozen invariant / artifact	How CC cites it
21	Six-stage pipeline spine const arrays byte-frozen to the Phase 21 baseline. Rest surface provider-purity (no /provider/i strings). The frozen Phase 21 test battery. Workflows 21A-21K closeout classifications (21E Operationally Validated).	Re-run the spine byte-freeze check and the Phase 21 battery at every closeout; UI work touching Rest/Working/Audit surfaces is additive only.
22	Voice Authority T0-T3 with the standing-consent store and revocation; voice is transport over the text pipeline; PTT runtime; TTS chains (demo-director: Chatterbox -> Kokoro -> Piper -> Edge-TTS -> captions, RECORDING-ONLY authority; PTT runtime: faster-whisper -> pipeline -> Piper -> playback, LIVE authority); openWakeWord wired, not listening; E-010 records the honest fallback status.	New voice work extends these tiers; it never redefines them. E-011 consolidates mechanism, never authority: 'shared mechanism, not shared authority'.
23	I1 single mutation path (resumeApproval sole executor; runtime.runTool = 2). I2 no raw frames/transcripts/OCR in any telemetry sink or view-model. I3 vision provider execution gated by default. I4 no background/autonomous capture. I5 pipeline/audit surfaces carry no execute/approve/mutate affordances. Multimodal packet schema rejects payload fields. duration_band additive. 23F webcam consent-gated with mandatory indicator; camera proof PENDING-HARDWARE.	PHASE23_CLOSEOUT_VERIFICATION.md is the evidence file; every later phase re-verifies I1-I5 with file:line.
24	JARVIS as a stdio MCP server, producer of proposals and reader of projections, never an executor. FC-1 server-derived canonical effect; FC-2 hash-frozen proposal re-validated at approval time; FC-3 identity from token, never body. GATE-1..5 (incl. the positive import allowlist and the unforgeable human-session marker). EoP-1..15 and ID-1..5 GREEN by drill. E-016/E-017 closed. Out of scope by declaration: consuming external MCP servers, HTTP/SSE transport, new mutation capability.	MCP_GATEWAY_THREAT_MODEL.md + MCP_GATEWAY_ARCHITECTURE.md. Any addition = exposure-matrix amendment + registry row + new slice. The 12-path non-bypass drill re-runs at every closeout that touches the gateway.

Phase	Frozen invariant / artifact	How CC cites it
Processes	The Enhancement Registry is the only entry for new work. Single agent, no subagents, no Agent-tool fan-out for building (parallel fan-out is allowed for read-only testing/probing only, with approval). Prompt-file workflow. Commits under standing authority only when every verification is green. No co-author trailers. Recon-before-brief.	Every CC prompt names its registry entry and states the single-agent rule in its role line.

4. JARVIS v5 End-State Architecture



Two things are drawn here that v4 left implicit. First, the Action Gateway is the existing frozen boundary (Phase 18 lifecycle wrapped by Phase 24 FC-1/FC-2), not a new component; ActionProposal v1 must extend the Phase 24 canonical proposal shape rather than introduce a second one. Second, the only two ways through the gateway are a human approval bound to a hash, or a Gate-minted mandate whose class/target/budget/expiry match. There is no third path, and Shadow is an evidence producer for the human, never an approver.

5. Complete Adoption Register

HOW TO READ THIS

'Adopt' means bring the principle or capability into the JARVIS target architecture. It does not mean copy source code or install every project. Each external system is a reference implementation or a replaceable driver behind JARVIS-owned contracts. **v5 adds a status check:** external facts were re-verified on 01 Sep 2026 and are re-verified again at adapter-entry time (the slice that adopts them), never trusted from this document alone. Where the runtime is alpha or high-churn the decision is 'pattern-adopt only' - the idea enters, the dependency does not.

Original JARVIS (keep, extend, upgrade)

Source	Part / pattern	Decision	JARVIS use
Original JARVIS	Capability-based model registry (Phase 13)	Keep	Provider/model names change; route by capability, latency, privacy and cost.
Original JARVIS	Four-stage router: intent -> safety -> capability -> cost	Keep + extend	Add evidence budget, runtime capability and behavioral reliability to routing (28B).
Original JARVIS	Latency budgets and streaming-first voice	Keep	Fast presence remains essential.
Original JARVIS	SQLite/event-store persistence + replay (Phase 11)	Keep + extend	Add durable worker/task/checkpoint and transition records. No second event store.
Original JARVIS	Approval lifecycle / trust tiers (Phase 18, 22)	Keep	Becomes the Action Gateway + Shadow; never replaced by prompt policy.
Original JARVIS	MCP Gateway (Phase 24) <i>(added)</i>	Keep, frozen	The server side of MCP is done and drilled. Consumer side is new (Track B2b).
Original JARVIS	Enhancement Registry <i>(added)</i>	Keep	Only entry for new work; every superseded decision is a registry amendment.
Original JARVIS	Plugin/adapter architecture	Keep	External runtimes become drivers/adapters.
Original JARVIS	Fake-first -> real-adapter conformance	Keep	New node/sensor/runner classes must pass parity tests.
Original JARVIS	Cost caps + loop budgets	Keep	Extend to per-worker, per-task and council evidence budgets.
Original JARVIS	Suggestion Inbox	Keep, narrow	User-facing proposals and the promotion inbox; durable workers may do permitted work inside mandates.
Original JARVIS	Voice Authority T0-T3 (Phase 22)	Keep, frozen	T0 direct read; T1 standing consent; T2 UI confirm; T3 manual.
Original JARVIS	Knowledge Compounding	Upgrade	Feed through provenance and the promotion firewall; the human promotes.
Original JARVIS	LLM Council (Phase 21 Council Mode)	Upgrade	Dynamic by risk/uncertainty, not 'more models = better' (28C).
Original JARVIS	Verification Agent (21A)	Upgrade	Integrate with outcome verification and Shadow evidence rather than advisory-only checks.
Original JARVIS	Telegram/mobile inlets (21)	Upgrade	Unified Channel Gateway with consistent identity/routing (30D).
Original JARVIS	Social media extraction (21E) + video pipeline (23)	Keep	Operationally validated local multimodal ingestion remains the sensor template.
Original JARVIS	WorkflowBox v1a-v1c <i>(added)</i>	Keep	Personal project surface; the amber door stays a type seam until the action layer is designed.

External (verified 01 Sep 2026)

Source	Part / pattern	Decision	JARVIS use
Grok Bot	Persistent specialist coworkers	Adopt	Builder/Researcher/Scout remain logically persistent while JARVIS is the single surface.
Grok Bot	Cloud computer model	Adopt as optional runtime	Cloud compute is borrowed execution, not JARVIS authority/home.
Grok Bot	Worker-to-worker messaging	Adopt	Durable typed peer bus with task IDs and provenance.
Grok Bot	Supervisor / chief-of-staff worker	Adopt carefully	Orchestrator coordinates specialists; no bot-clutter on the user surface.
Grok Bot	Teach-a-task / watch-me	Adopt	Demonstrations become Procedure Candidates, then replay/verify, then human promotion.
Grok Bot	Learn working preferences	Adopt	Structured Worker/User Profile, not hidden fine-tuning.
Grok Bot	Proactive workers / standing mandates	Adopt, Gate-minted	Only inside a MandateGrant v1 minted through the approval lifecycle; scheduler wakes, never authorizes.
Grok Bot	Loose-ends recovery	Adopt	Unfinished tasks become durable unresolved work.
Grok Bot	Phone supervision / presence states	Adopt	Approve, steer, inspect, take wheel, resume from phone; working/waiting/blocked/sleeping/offline states.
OpenBot (CopilotKit)	Single action gateway; audit-before-act; deny-before-allow; take-the-wheel; secrets absent from transcript; shell through the gateway; private loopback endpoints	Adopt pattern P0	First tag v0.0.1 on 17 Aug 2026, self-described alpha: no runtime dependency . The patterns map onto the existing Gate; policy compiled, fail-closed, deny evaluated first.
OpenBot	Per-worker governed computer; supervisor owns container creation	Adopt where needed	Isolated browser/workspace per specialist; worker cannot control its own security substrate.
OpenBot	Governed MCP	Adopt pattern, consumer side only	Server side exists (Phase 24). Unknown effects default to write/riskier classification when JARVIS calls out (B2b).
OpenBot	Generative components instead of prose-only	Adopt	Health, approvals, traces, tasks and world state render as typed components (U.7).
OpenClaw	Single local Gateway control plane; multi-channel inbox; node abstraction (P0); node-hosted skills; persistent scheduler outside model (P0); webhooks/wake events; voice wake sync; sandbox non-main workers; managed skills	Adopt pattern	Channels, sessions, nodes and events converge through one JARVIS gateway. Scheduler pattern lands as a Phase 8 wrap (section 23).
Hermes Agent	Durable bot-to-bot peer messaging (P0); persistent cron memory; change-suppressed monitors; live steer/stop subagents (P0); structured subagent outputs (P0); per-subagent cost; profile isolation; completion wakeups; skill creation from experience (behind firewall); VPS portability	Adopt pattern	v0.21.0 released 31 Aug 2026 (verified); ~5,800 commits since v0.20 - high churn, pattern-adopt only. Structured outputs and steer/stop land in 28D and Track A3.
AG-UI	Standard agent-user event protocol; bidirectional state sync; generative UI with allowlist	Evaluate / adopt adapter	Useful for streaming, state sync, components, interrupts; app decides what mounts/executes (B6).

Source	Part / pattern	Decision	JARVIS use
NVIDIA OpenS hell/NemoClaw	Out-of-process sandbox enforcement (P0); deny-by-default + live policy update; privacy router	Adopt principle	Security control lives outside the worker/model process; versioned policy is runtime config, not prompt text.
AWS AgentCore	Serverless microVM runtime; runtime instances up to 14 days; persistent session storage; GPU driver	Evaluate (E-track)	Runtime instances GA Aug 2026 (verified). Optional cloud driver for long jobs; session-state abstraction adopted regardless of vendor.
RuView	Unified spatial world model (P0 concept); hardware adapter registry; age/geome try/uncertainty-aware observations (P0); RF spatial memory (observe-only); synthetic RF worlds (evaluate); active sensing (defer)	Adopt concept / observe-only	One persistent representation; every observation carries time, frame, confidence, provenance. No physical truth until real-world validation (27G, 29G).
Shadow/Savepoint (own research)	Intent Contract (P0); deterministic counterfactual preview (P0 for software); Transition Record (P0); process != outcome (P0); state adapters; evidence budget; post-state verification; recovery as evidence; fidelity score; supervisor-held approval secret; default-deny network in forks; harm-vs-autonomy benchmark	Adopt	Track C. Intent Contract is an input to ActionProposal; the approval still binds the FC-2 hash. Shadow evidences; it never approves.
Sleepwalk/Lucid (dissertation)	Provenance-tagged memories (P0); quarantine for untrusted candidates (P0); behavioral promotion gate; activation drift monitoring (research-only)	Adopt, human-promoted	Automation quarantines, scores, replays, canaries, proposes. A human promotes through the Gate (Track D).
AffectiveSec (own research)	Behavioral stress evaluation	Adopt in model qualification	Repeated failure, pressure, contradictions, poisoned docs before any model is assigned authority (28F).
Soup	Local adapter/LoRA training driver (evaluate later); immutable plan -> confirm -> execute binding; SHIP / DON'T SHIP gate	Adopt principle	Approval binds the exact operation/snapshot - this is FC-2, already frozen; cite it.
Human Operator	Bidirectional physical guidance (long-term R&D); symbolic motor intent separated from actuation; physical emergency stop	R&D only / mandatory if pursued	Track G. Hardware-level fail-off outside model/software authority.
DIY JARVIS community	Fast conversational brain + deep background brain	Adopt P0	Presence answers immediately while worker runs continue asynchronously (26B).

6. Canonical Runtime Contracts

Contracts are frozen by slice, versioned, and reuse what exists. The 'Reuse' column is the audit's checklist: if an existing store or schema satisfies the contract, extend it; do not create a parallel one (v4 section 22 rule, now per-contract).

Contract	Purpose / required fields	Reuse / constraint
WorkerProfile v1	worker_id, role, principal, model policy, memory namespace, tool grants, mandate refs, runtime requirements, status	Principal is always the single JARVIS principal; tool grants can only narrow.
TaskEnvelope v1	task_id, parent_id, origin, intent_ref, worker, priority, deadline, retry budget, checkpoint policy	Persist in the Phase 11 event store; scheduled origin comes from the Phase 8 scheduler (wrapped, section 23).
IntentContract v1	goal, success criteria, forbidden outcomes, evidence budget, cost/time budget, approval policy	Input to ActionProposal. Never a second approval object.
WorkerMessage v1	from, to, task_id, type, artifact refs, payload summary, requires_reply, expiry, provenance	Payload summary only; message bodies are DATA, never instructions (EoP-11).
StructuredResult v1	typed ResearchResult / BuildResult / ReviewResult / ObservationResult	Malformed results rejected or quarantined (28D).
NodeCapability v1	node_id, platform, online state, capabilities, permissions, locality, accelerator, sensors, skill inventory	A node advertises; it never becomes the worker or raises its mandate.
ExecutionLease v1	task/run, node, capability set, expiry, policy version, intent ref, checkpoint ref	Lease expiry = hard stop. No lease, no execution.
ActionProposal v1	actor, target, semantic action, normalized args hash, authority tier, sensitivity, reversibility	Must extend the Phase 24 canonical proposal (FC-1 server-derived effect, FC-2 hash, scope snapshot, expiry). One proposal shape in the repo.
MandateGrant v1 (new)	mandate_id, principal, worker_id, action_class, target_class, budget (count/cost/time), expiry, approval_ref (FC-2 hash of the mandate proposal), revocation state, receipt refs	Minted only via the approval lifecycle as a T2 UI-confirm. Not creatable, widenable or renewable by any worker, scheduler or node. Checked at execution time, not issue time.
TransitionRecord v1	pre-state, proposed action, evidence, decision, actual post-state, process/outcome verification, privacy	Append-only; pairs with the existing audit event rather than replacing it.
SpatialObservation v1	sensor, canonical frame, timestamp/age, value, uncertainty/confidence, provenance, fidelity	Frozen at 27A; all sensors map to it.
WorldState v1	canonical coordinate frame, entities, devices, bodies, virtual objects, confidence-weighted state	One store; renderers consume, never author.
Procedure v1	intent, inputs, preconditions, steps, verification, recovery, risk tier, provenance, version, success stats	Executable knowledge = T2 at minimum; promotion is human.
PromotionDecision v1	candidate ref, provenance score, replay result, canary result, conflicts, human decision , promote/quarantine	'Reviewer' is a human-session marker (GATE-4 pattern), never a model id.
Receipt v1	who/what/when/where, policy decision, approval or mandate ref, evidence refs, post-state verification, hashes	Emitted for every T1/T2 execution; the existing audit event plus a TransitionRecord ref.

7. Roadmap Structure and Sequencing Rule

DO NOT RENUMBER HISTORY - AND DO NOT REUSE ITS NUMBERS

Phases 1-24 are historical truth with the scopes below. v4 assigned 22-26 to new work; those numbers are taken. v5 assigns 26-30 to the five capability phases, **provisionally**: audit step 0 confirms no phase or registry entry already claims a number before any slice opens. If the audit finds a collision, the collision halts the slice and the number is resolved in the registry first. Tracks A-G stay unnumbered until the audit maps them onto the repository.

Number	Real scope (repository)	Status	v4 called it
1-20	Governance and operationalization substrate	FROZEN 31 May 2026	same
21	Expansion workflows 21A-21K; cockpit surfaces; pipeline spine	FROZEN 03 Jun 2026	same
22	Voice Authority T0-T3, standing consent, PTT/TTS runtime, wake word wired	FROZEN (on main via E-001/E-006; fallback truth in E-010)	'Phase 22 Presence & Voice Overhaul' (future)
23	Vision + Sensing Realization 23A-23G, invariants I1-I5	FROZEN (23F camera proof PENDING-HARDWARE)	'Phase 23 Vision, Sensor Fusion & Spatial World Model' (future)
24	MCP Gateway 24A-24E, non-bypass drill, E-016/E-017 closed	FROZEN, zero residuals	'Phase 24 Local Intelligence, Council' (future)
Runway	Audit, stabilize, E-011, real-vs-synthetic check	NEXT	(absent; U started immediately)
Program U	Command Center capstone (UI_CAPSTONE_BRIEF.md)	after runway; closes Execution Enabled	'Program U' (same, closes OV)
25	Productionization : synthetic-to-real, voice live, bootstrap, packaging, soak	after U; gate OPEN (move confirmed 01 Sep 2026)	(absent)
26	Presence & Voice Overhaul	provisional	22
27	Vision, Sensor Fusion & Spatial World Model	provisional	23
28	Local Intelligence, Council & Model Qualification	provisional	24
29	Smart Room, Sensing & Spatial Interaction	provisional	25
30	Mobile JARVIS & Node Fabric	provisional	26
Tracks A-G	Durable workers, governed computer, Shadow, memory trust, cloud continuity, spatial gauntlet, augmentation R&D	unnumbered until audit	same

8. Pre-Phase-25 Runway (new in v5)

Four bounded steps that the July session sequenced and v4 dropped. None is a phase; each is a registry-tracked slice with its own closeout. R.1 and R.2 are mandatory before anything else in this document; R.3 and R.4 are mandatory before Program U.

Slice	Execute	Exit / DONE
R.1 Repository reality audit	Read-only, one report file (docs/audits/V5_ENTRY_AUDIT.md). Map current main to sections 3, 6 and 7 of this document. Classify every capability with the status vocabulary. Confirm: frozen phase list and numbers; registry state of E-005/E-009/E-011/E-012/23F; runtime.runTool = 2; spine byte-freeze; which machine is present; whether Phase 25 has been opened anywhere. Locate the approval lifecycle, scheduler, router, registry, event store, plugin system, voice stack, vision stack, room adapters, Council, Telegram, Google/MCP, UI surfaces, disabled-feature guards. No fixes.	Report cites file:line for every claim; numbering-collision check returns CLEAR or names the collision; no implementation mixed in.
R.2 Stabilize	Green baseline on the primary machine: dependencies, target bootstrap, the E-013/E-015 flake family (raise timeouts via registry entries; the whole-repo-scan-in-one-it shape is banned), gauntlet and JARVIS suites run separately.	Full suite green in-hook, no --no-verify, three consecutive clean runs.
R.3 E-011 voice surface consolidation	Execute the written prompt (e011-voice-consolidation.md) after re-checking it against the fresh tree. Extract ONE canonical TTS engine + failover layer that both the demo-director chain (recording-only) and the PTT runtime (live) call. Re-point the Phase 23 Piper failover drill to the shared layer with identical assertions.	Drill green; recording-only vs live authority boundaries provably separate ('shared mechanism, not shared authority'); zero new mutation paths; E-011 COMPLETE.
R.4 Real-vs-synthetic check (light)	Sweep every cockpit-visible value and classify LIVE vs DELIBERATELY-SYNTHETIC (a fail-closed boundary) vs SHOULD-BE-LIVE. Output one table. The full realization is Phase 25A/25B; this only tells the capstone what it may express.	Table committed under docs/audits; every synthetic value the capstone will render has an honest inline label decided here.
R.5 23F camera proof (any time)	When a webcam is present: enumerate -> consent -> indicator -> capture -> kill-drill.	PENDING-HARDWARE -> VERIFIED in the registry. Five minutes; the last Phase 23 residual.

9. Program U - Command Center Capstone (before Phase 25)

WHAT CHANGED FROM v4, AND WHY

Program U is the July 'UI capstone' under v4's name. It now cites its brief and the frozen Phase 21 surfaces, U.6 extends the Phase 21 orb instead of rebuilding it, motion is Framer Motion written from official docs (the unvetted 0-star skill stays rejected), and U.8 moves to Phase 25 because operational validation needs real data.

Slice	Execute	Exit / DONE
U.1 Reality audit	Audit current UI against DESIGN.md, PRODUCT.md, UI_CAPSTONE_BRIEF.md and the R.4 table. Screenshot inventory + dead/legacy component map. Confirm the cockpit grammar: SVG not canvas; Human Gate central; AMBER = Gate-touching, CALM = read-only; trusted/untrusted visual separation (EoP-11); --jarvis-* tokens (Syne + JetBrains Mono).	Audit cites files/components; no implementation mixed into audit.
U.2 Design tokens + typography	Centralize semantic color/spacing/type/motion tokens; remove Arial/generic overrides; no invented tokens.	Zero hard-coded rogue colors in target surfaces; a11y contrast passes; no purple/blue startup gradients.
U.3 Chatbot removal	Replace chat-centric shell with command/presence/work surface while preserving conversation as one pane. Rest surface keeps provider-purity.	No legacy 'Message JARVIS / thinking bubble / provider dropdown' as primary UX; Phase 21 battery and Rest /provider/i purity still green.
U.4 Cockpit layout	Operator rails: Presence, Work (WorkflowBox lane + mind-map), World, Power/Cost, Approvals, Evidence.	Responsive desktop layout; no card-soup; keyboard navigation passes; I5 affordance invariants green.
U.5 Motion/state language	Framer Motion from official docs; motion tokens and reduced-motion path; states idle/listening/working/blocked/approval/error.	Motion deterministic, interruptible, accessible.
U.6 Orb state machine	Extend the Phase 21 orb (seven states) to reflect real runtime states and voice amplitude from the E-011 consolidated layer. CSS/Three.js decision from the audit; do not rebuild.	Orb reflects real runtime states; no fake decorative states; Phase 21 orb tests remain green.
U.7 Generative components	Typed components for system health, task progress, approval (canonical effect, never client framing), worker status, world state, trace summary. 21st.dev Magic output is conformed to the grammar before shipping.	Components render from typed schemas; no raw code/payload leakage; every synthetic feed labelled per R.4.
U.8 -> moved to 25F	E2E + performance closeout runs against real API/event data, which exists only after Phase 25A-25D.	Program U closes at Execution Enabled . Operationally Validated is asserted by 25F.

10. Phase 25 - Productionization (new in v5; the DONE phase)

WHAT CHANGED FROM v4, AND WHY

Absent from v4. Defined in the July session as 'audit -> realize each with a drill -> wake word live -> package', gated behind the move - which has happened, so the gate is OPEN. It comes before any new capability phase because it is the only phase whose closeout makes the words 'target hardware' and 'real integration' checkable for every phase after it. Order is risk-ascending: classify before touching; realize one value at a time; package last, because packaging synthetic-data JARVIS ships the aspiration in a nicer wrapper.

Slice	Execute	Exit / DONE
25A Synthetic-to-real audit	Read-only. Catalogue every synthetic/fixture value in runtime and UI. Classify each as LIVE-INTENDED or DELIBERATELY-SYNTHETIC (fail-closed boundary, e.g. gated providers, blocked CAI). Rank LIVE-INTENDED by risk. One report.	Every value classified with a reason; the DELIBERATELY-SYNTHETIC list is explicitly preserved as governance, not debt.
25B Realize with a drill	For each LIVE-INTENDED value, in risk order, one at a time: wire the real source, then a kill-drill (the Piper failover discipline). No big-bang.	Each realization has its own drill test and registry row; zero new mutation paths; suite green after each.
25C Voice stack live	openWakeWord listens on the primary machine: wake -> visible mic state -> one governed capture (PTT-equivalent session) -> text pipeline. No cloud audio before wake; no standby audio stored. Frozen T0-T3 enforced. E-012 real fallback beyond demo-director closes here through the E-011 layer.	Wake/false-wake/permission tests on real hardware; failover drill green on both chains; T2/T3 voice bypass fails closed; E-012 COMPLETE.
25D Target-hardware bootstrap	Fresh clone -> bootstrap -> Ollama + Piper + faster-whisper + Chatterbox sidecars (Apple Silicon builds; MLX voice stays a later option) -> keys (DeepSeek, Google OAuth) -> Obsidian vault path -> suite green, on the M1 Max 32GB. This is JARVIS's first run on the Mac; document every manual step and every Windows-path assumption found.	A second person could bring JARVIS up from the README on an Apple Silicon Mac; the machine and its measured envelope are recorded in the registry.
25E 23F camera proof	If not already done in R.5.	PENDING-HARDWARE -> VERIFIED.
25F UI operational validation (was U.8)	Playwright/a11y/perf against real API/event data on the primary machine; every critical journey; every R.4 SHOULD-BE-LIVE value now live or explicitly deferred.	Program U flips to Operationally Validated.
25G Packaging	Bundle Next.js server + Node + Python sidecars + SQLite into one launchable app (Tauri shell, 127.0.0.1 only, sidecar process management, first-run bootstrap). Last, by design.	Cold launch from the packaged artifact on the primary machine reaches a working cockpit and a voice round-trip; no public network exposure.
25H Soak closeout	Seven consecutive days of real daily use from the packaged build: receipts, telemetry and audit prove it. Chaos: kill process, expire approval, deny permission, drop a sidecar.	Phase 25 Operationally Validated; 'production-grade' may now be claimed for the daily-driver core. Every residual closed or hardware-gated by name.

11. Phase 26 - Presence & Voice Overhaul (v4: Phase 22)

WHAT CHANGED FROM v4, AND WHY

22E (voice authority T0-T3) deleted - frozen Phase 22. 22C (wake word) moved to 25C. 22G (voice-node protocol) moved to 30F. What remains is genuinely new: the fast-brain/deep-worker split, the multi-turn conversation state machine, and duplex streaming, all measured on the real machine.

Slice	Execute	Exit / DONE
26A Target-hardware voice benchmark	Benchmark current STT/TTS options on the primary machine through the E-011 layer; first-token, end-to-end, memory, CPU/GPU, interruption quality. Registry result, not preference.	Chosen stack meets target P50/P95 and fallback works.
26B Presence Engine	Separate the ultra-fast conversational path from the deep-worker path. Fast brain handles acknowledgements/simple conversation; deep tasks become durable jobs (TaskEnvelope).	Simple speech response starts within target; complex task acknowledges immediately then continues asynchronously.
26C Conversation state machine	Sleep -> Wake (25C) -> Conversation Idle/Active -> Sleep with VAD, timeout, explicit sleep, barge-in.	Repeated live sessions survive interruptions and clean teardown; no raw standby audio stored.
26D Streaming duplex voice	Sentence/chunk streaming, audio queue, user interruption, fallback voice.	No full-response wait; queue flushes on barge-in; fallback smoke passes.
26E Operational closeout	Live run: wake -> converse -> T0 query -> T1 consented room action -> T2 proposal -> UI confirm -> barge-in.	All authority, latency, privacy and failure gates pass; I1-I5 re-verified.

12. Phase 27 - Vision, Sensor Fusion & Spatial World Model (v4: Phase 23)

WHAT CHANGED FROM v4, AND WHY

Renumbered to avoid the frozen Phase 23. 27B extends the 23F consent-gated webcam and the Phase 23 packet pipeline; 27C reuses the Phase 15 screenshot-OCR sensor rather than building a new one. Nothing here touches I1-I5 except to re-verify them.

Slice	Execute	Exit / DONE
27A Sensor contract	Freeze SpatialObservation v1 and canonical coordinate-frame/provenance rules.	All current/future sensors map without bespoke world-state paths.
27B Real RGB/depth capture	Extend 23F: camera/depth behind the consent indicator; local object/hand/pose/OCR first, cloud fallback explicit.	Live frames produce typed observations; I2/I4 privacy tests prove frames are never generically logged.
27C Screen perception	Phase 15 screenshot/OCR/app context re-exposed as a sensor adapter with explicit user-visible capture state.	Real desktop perception smoke; protected apps/areas can be excluded.
27D Sensor fusion engine	Fuse by time, geometry, source reliability and uncertainty; contradictions remain explicit.	Synthetic and live disagreement tests produce bounded confidence, never false certainty.
27E Spatial World Model	Persist room entities, bodies/hands, devices, virtual objects, confidence-weighted state.	World state survives restart and updates incrementally from live sensors.
27F Spatial query API	'Where is X / who is present / what moved' read canonical world state, not individual sensors.	UI/voice consume one API and show source confidence.
27G RuView experimental adapter	Observe-only adapter; record evidence level/fidelity; never sole high-risk authority.	Real hardware smoke if hardware exists; otherwise 27G stays PENDING-HARDWARE and cannot close.

Slice	Execute	Exit / DONE
27H World-model closeout	Real demo: RGB/depth + one non-vision sensor into one state; recover from sensor loss.	Spatial capability Operationally Validated.

13. Phase 28 - Local Intelligence, Council & Model Qualification (v4: Phase 24)

WHAT CHANGED FROM v4, AND WHY

Renumbered to avoid the frozen Phase 24. 28A benchmarks the confirmed M1 Max 32GB (the July 18GB envelope is stale and is re-measured, not assumed). 28C upgrades Phase 21 Council Mode rather than inventing a council. v4's 24E 'live worker steering' depended on a worker runtime that does not exist until Track A - moved to A3; 28E keeps only what today's runner can do (cancel/stop with cost preview).

Slice	Execute	Exit / DONE
28A Primary-machine capability benchmark	Benchmark local models by task class on the M1 Max 32GB: conversation, classification, code, vision, structured output, tool planning. Runs early as a measurement-only dependency of 26A. Prior to test: 8B workhorse; 14B comfortable; 27-32B-class at 4-bit; 70B out.	Registry contains measured latency/RAM/quality for the 32GB envelope; the July 18GB assumptions are retired; fit scorer thresholds updated from measurement.
28B Router v2	Add hardware locality, worker reliability, behavioral stress score and evidence budget to the Phase 13 capability/cost router.	Router decisions are explainable and reproduce benchmark-backed choices.
28C Dynamic Council	Upgrade Phase 21 Council Mode: invoked by risk/uncertainty/topology; coordinator -> specialists -> reviewer, not fixed fan-out; cost gate retained.	Simple tasks never burn council; high-risk task escalates with cost preview.
28D Structured worker outputs	Enforce StructuredResult v1 schemas.	Malformed/free-form results are rejected or quarantined.
28E Run control	Cancel/stop/cost-stop for council and long runs using existing kill semantics. Steer/pause/resume/checkpoint is Track A3.	User can stop a run mid-flight without losing completed evidence.
28F Behavioral stress qualification	AffectiveSec-inspired scenarios: repeated failure, time pressure, contradictions, poisoned docs, tool outages.	Registry records safety/reliability under stress, not benchmark scores alone.
28G Local/cloud fallback	Provider failure triggers explicit fallback (Phase 13 fail-closed registry) without changing worker identity or authority.	Outage smoke completes via approved alternative and logs reason; cloud never enabled by fallback alone.
28H Closeout	Representative local-first task plus council escalation with measured cost, latency, structured handoffs and stress-qualified models.	Routing/council stack Operationally Validated.

14. Phase 29 - Smart Room, Sensing & Spatial Interaction (v4: Phase 25)

WHAT CHANGED FROM v4, AND WHY

Renumbered. 29B is hardware-gated by name: no real device on the registry means 29B cannot close, and the phase cannot claim room control from fake adapters (frozen: fake-first, real devices one at a time, observe_only default).

Slice	Execute	Exit / DONE
29A Device registry v2	Extend the Phase 6/16 registry: lights/plugs/audio/display/sensors with trust class and verification-read support.	No device class bypasses adapter conformance.
29B Real room adapter validation	Smoke the real Hue (or whichever device exists) on current hardware; retire stale fake-only claims.	Real read -> dry-run -> approval -> write -> verification read proven, or PENDING-HARDWARE by device name.
29C World-state room actions	Room actions query the Spatial World Model and current device state before execution.	Conflicting/stale state fails closed or asks.

Slice	Execute	Exit / DONE
29D Spatial audio	Map audio source positions to the canonical room frame; directional cues and state tones.	Audible spatial movement follows world-state coordinates reliably.
29E Gesture interaction	Depth/RGB/IMU hand tracking maps gestures to semantic commands against virtual/physical objects.	False gesture rate benchmarked; a gesture can never directly authorize a T2/T3 action.
29F Haptic feedback	Safe low-voltage haptics for selection, collision, confirmation, direction.	Event-driven and bounded; emergency stop tested.
29G RuView corroboration	Compare RF observations against depth/RGB/mmWave ground truth; assign trust from measured results.	No promotion above observe-only without the predefined accuracy/reliability threshold.
29H Closeout	Live one-room demo: presence + gesture + room state + spatial audio/haptics + verified device side effect.	Room-spatial loop Operationally Validated.

15. Phase 30 - Mobile JARVIS & Node Fabric (v4: Phase 26)

WHAT CHANGED FROM v4, AND WHY

Renumbered. 30D extends the Phase 21 Telegram inbound transport. 30F absorbs v4's 22G voice-node protocol because its exit criterion needs the phone inlet to exist.

Slice	Execute	Exit / DONE
30A Node protocol	Freeze NodeCapability/Heartbeat/Invoke/Result/Lease contracts. Mac, PC, phone and future devices are capabilities, not special-case code paths.	At least Mac + Windows register and update availability/capabilities.
30B Windows execution node	Governed node on the ASUS for Windows/Unreal/GPU tasks; outbound trusted connection; no raw remote shell exposure.	Mac assigns a safe test job; Windows executes through the gateway and returns artifact + receipt.
30C iPhone companion	Tauri/native companion for command, worker status, approvals, notifications, take-wheel and sensors.	Real device smoke over local/Tailscale transport; approvals carry the human-session marker (GATE-4).
30D Channel Gateway	Unify Telegram (Phase 21) + phone + desktop inbound routing and outbound notifications.	Same task identity/authority semantics across channels.
30E Take the Wheel	Live stream/control handoff for browser/desktop tasks needing human login/2FA.	help_requested -> control_taken -> control_released verified; worker blocked while the human drives; secret absent from transcript.
30F Cross-device voice/wake + voice-node protocol	Gateway-owned wake/presence config, local detection, explicit device capability; voice events normalized so any allowed inlet carries the same authority metadata.	Same utterance produces the same authority decision regardless of inlet; no cloud audio capture.
30G Offline/degraded modes	Explicit local/Tailscale/cloud/offline state; never silently degrade authority or privacy.	Disconnect/reconnect/resume tests pass.
30H Closeout	Start task on Mac, inspect/steer on phone, execute a Windows-specific subtask, survive one node disconnect, complete with receipts.	Node fabric Operationally Validated.

16. Post-Phase-30 Expansion Tracks (Unnumbered Until Repo Reconciliation)

WHY UNNUMBERED

Approved target capabilities from the research synthesis. They are numbered only after the R.1 audit maps them onto the repository. v5 edits: Track A reuses the Phase 8 scheduler and Phase 11 store; Track B gains the governed MCP *client* (B2b) with a Phase-24-method threat model; Track C makes the Intent Contract an input to the proposal, never a second approver; Track D makes promotion human; A3 absorbs live worker steering from v4's 24E.

Track A - Durable Worker Runtime

Slice	Execute
A1 Worker aggregate	WorkerProfile v1 with model-independent identity, memory namespace, mandate refs, capabilities, status, preferred/fallback brains.
A2 Durable task store	TaskEnvelope, checkpoints, attempts, dependencies, blocked/waiting state and loose ends persisted in the Phase 11 event store, outside model context. No second store.
A3 Worker lifecycle	start/pause/resume/steer/cancel/checkpoint/handoff/escalate with idempotency (absorbs v4 24E live steering).
A4 Durable peer bus	Typed WorkerMessage, acknowledgements, expiries, replies, task-linked history. Bodies are data, never instructions.
A5 Recurring worker memory	Persistent scratchpads and change-suppressed monitors; the Phase 8 scheduler (wrapped per section 23) remains the source of time truth.
A6 Presence	working/waiting/blocked/offline/sleeping exposed to the Command Center and mobile.
A7 Operational proof	Multi-hour worker task survives process restart and resumes from checkpoint with the same identity, intent and mandate; every T1 execution under the mandate produced a Receipt.

Track B - Governed Computer & Browser

Slice	Execute
B1 Computer supervisor	Per-worker isolated browser/workspace runtime; supervisor owns lifecycle and low-level tokens (OpenBot pattern; no OpenBot runtime dependency while alpha).
B2 Browser/file/shell gateway	Every effect returns to the central Action Gateway; no alternate endpoint used by workers.
B2b Governed MCP client (new)	JARVIS consuming external MCP servers: allowlisted servers, tool effects classified server-side (unknown = write/riskier), every call an ActionProposal, results are untrusted data (EoP-11). Threat-modelled STRIDE-by-EoP exactly as Phase 24; the 24 server stays untouched.
B3 Fail-closed policy	Deny before allow; malformed/missing policy blocks.
B4 Credential locality	Credentials runtime-local/write-only; secrets never enter transcript or generic logs.
B5 Human handoff	Take-the-Wheel with audited control transfer and worker lockout (30E).
B6 AG-UI adapter	Evaluate for streaming state, generative components, interrupts and bring-your-own-agent endpoints, behind an allowlist.
B7 Operational proof	Worker completes a real browser workflow, hits 2FA, yields to the human, resumes and produces a verified artifact.

Track C - Shadow Consequence Layer

Slice	Execute
C1 Intent Contract	All consequential tasks carry immutable success/forbidden/budget/approval semantics as input to ActionProposal.

Slice	Execute
C2 Action normalization	Semantic action representation and stable action hash before policy/approval - this IS FC-1/FC-2 generalized beyond MCP; reuse the Phase 24 canonicalization leaf.
C3 Savepoint integration	Shadow/Savepoint for coding/git/filesystem shell where deterministic forks are available.
C4 State adapters	Git/filesystem/browser/database/room/world pre/post state adapters.
C5 Outcome verifier	Deterministic validators first; model verifier only where needed; process_ok separate from outcome_ok.
C6 Evidence allocator	Risk x uncertainty chooses one check, deeper simulation, reviewer/council, or human escalation. It can raise the bar; it can never lower it below the tier's default.
C7 Transition Record	Append-only paired pre/action/evidence/decision/post/outcome record with privacy controls.
C8 Recovery policy	Blocked/failed actions produce safer alternatives while preserving progress.
C9 Benchmark	Vanilla vs governed vs Shadow on damaging transitions, completion, false blocks, approvals, latency and recovery.
C10 Operational proof	Real coding workflow runs for an extended period with lower approval spam and verified final state. Shadow never approved anything; the human or a mandate did.

Track D - Memory Trust & Procedure Learning

Slice	Execute
D1 Provenance everywhere	Every durable fact/episode/experience/procedure candidate has source, timestamp, trust and derivation.
D2 Quarantine store	Unknown/untrusted candidates cannot enter canonical memory automatically.
D3 Demonstration capture	Watch-me-once workflow capture generates a candidate procedure, not executable automation.
D4 Replay/verification	Candidate runs in a safe environment against the Intent Contract and expected post-state.
D5 Behavioral canaries	Promotion checks for capability regression, policy escape and contradictory behaviors.
D6 Procedure registry	Versioned name/inputs/preconditions/steps/verifier/recovery/risk/provenance/success statistics.
D7 Experience compiler	Verified transition trajectories distilled into candidate procedures; failures also teach recovery.
D8 Promotion inbox (new)	Batched human promotion through the Gate: candidate + provenance score + replay + canary results + conflicts rendered as canonical effect; the human promotes or quarantines; the decision carries a human-session marker.
D9 Operational proof	Teach a real repeated workflow once, verify it, a human promotes it, then it repeats successfully from one short instruction. The Trust demo (section 26) passes.

Track E - Cloud Continuity Driver

Slice	Execute
E1 RuntimeDriver interface	LocalNode / AgentCore / other cloud runtime behind the same task/lease/checkpoint contract.
E2 Placement policy	Choose node by capability, privacy, cost, latency, availability and credential locality.
E3 Checkpoint portability	Task survives runtime process loss; logical identity and intent unchanged.
E4 Cloud borrowed compute	Long-running browser/research/render jobs may continue while the Mac sleeps, subject to explicit cloud policy; any T2 blocks on phone approval (30C); secrets never leave the local node.
E5 Local return	When local-only capability returns, task checkpoints/hands back rather than restarting.
E6 Operational proof	Start local -> local offline -> approved cloud-safe portion continues -> resume local-only finalization.

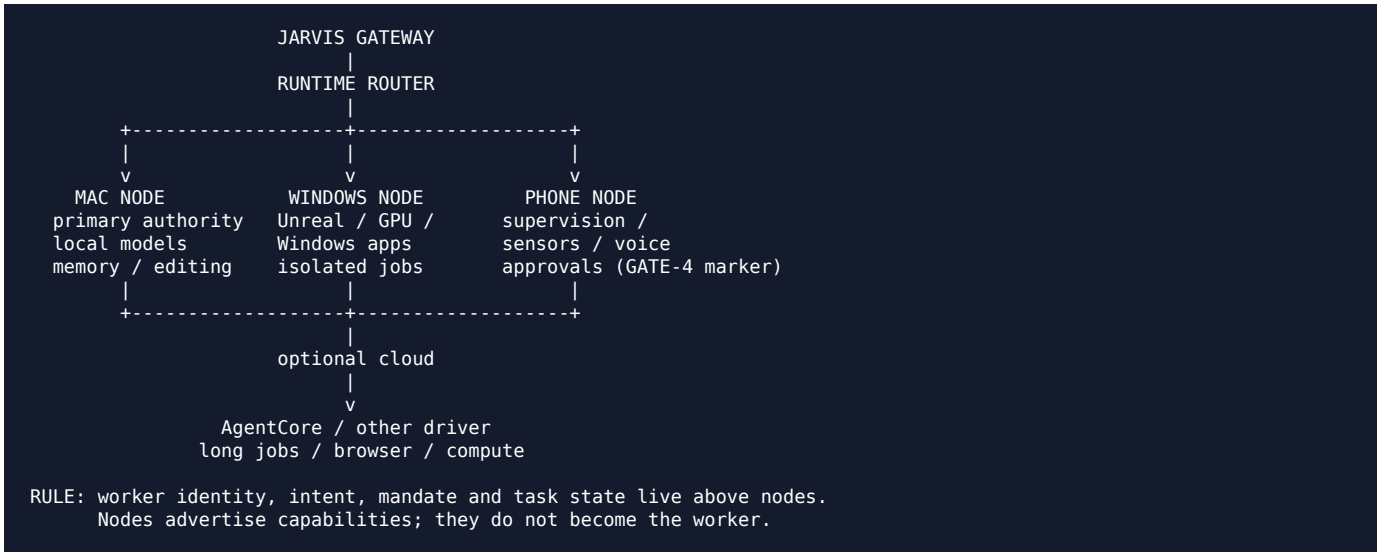
Track F - Spatial Interface & Gauntlet

Slice	Execute
F1 Virtual-object schema	Objects in canonical world coordinates with ownership, state and interaction affordances.
F2 Glasses/render adapter	AR/spatial display consumes WorldState; Unreal may be renderer, not source of truth.
F3 Hand ray/pinch/manipulation	Semantic gestures manipulate world objects with confidence thresholds and cancelability.
F4 Cross-device spatial sync	Mac/Windows/glasses/phone observe the same object state.
F5 Spatial audio/haptics	Feedback moves with objects/events, not canned effects.
F6 Shadow for physical actions	Physical/room actions use fidelity-aware state and the consequence gate before side effects.
F7 Operational proof	Manipulate a virtual gauntlet/object with a tracked hand, synced visuals + audio/haptics, world state coherent.

Track G - Human Augmentation Research (R&D only)

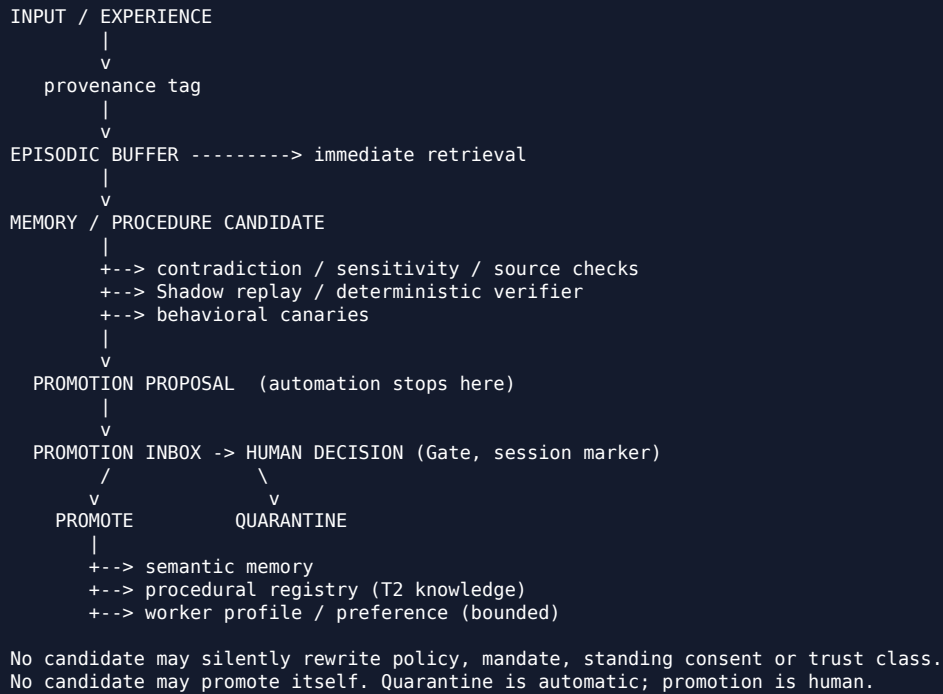
Slice	Execute
G1 Bidirectional interface research	Gauntlet as potential output channel as well as input.
G2 Symbolic motor intent	If physical guidance is ever tested, the model emits a symbolic target only; a deterministic controller owns actuation parameters.
G3 Independent hardware watchdog	Physical disconnect, current/duration bounds and emergency stop outside model authority.
G4 Ethics/medical/safety review	No EMS experimentation as product work without proper supervision, informed consent and expert review.
G5 Kill criterion	If safe bounded guidance cannot be independently enforced, do not pursue.

17. Multi-Machine Execution Architecture



Placement input	Meaning
Capability	OS, GPU, browser, Unreal, local files, sensors, model runtime, camera, audio.
Privacy	Data class may prohibit cloud or a specific node.
Credentials	Prefer action where the secret already lives; pass capability, not raw secret.
Latency	Conversation stays local/fast; long jobs may tolerate remote.
Cost	Cloud only when local capacity/availability/latency cannot satisfy the task.
Authority	A node cannot increase a worker's mandate. Approvals from a node carry the human-session marker or are rejected.
Availability	Offline node causes checkpoint/wait/handoff, never silent task disappearance.
Fidelity	Execution/simulation location declares what external state it cannot reproduce.

18. Memory & Learning Architecture v5



Why the human is in the loop here (v5 change 14). The frozen substrate says 'no silent writes' and 'derived state is not canonical truth'. A promoted procedure is executable knowledge that a later short instruction will run; a promoted fact shapes every future answer. Letting a model reviewer promote either is precisely the sleep-time consolidation attack surface the Sleepwalk/Lucid work studies. The inbox is batched so the cost is one review session, not an approval per candidate.

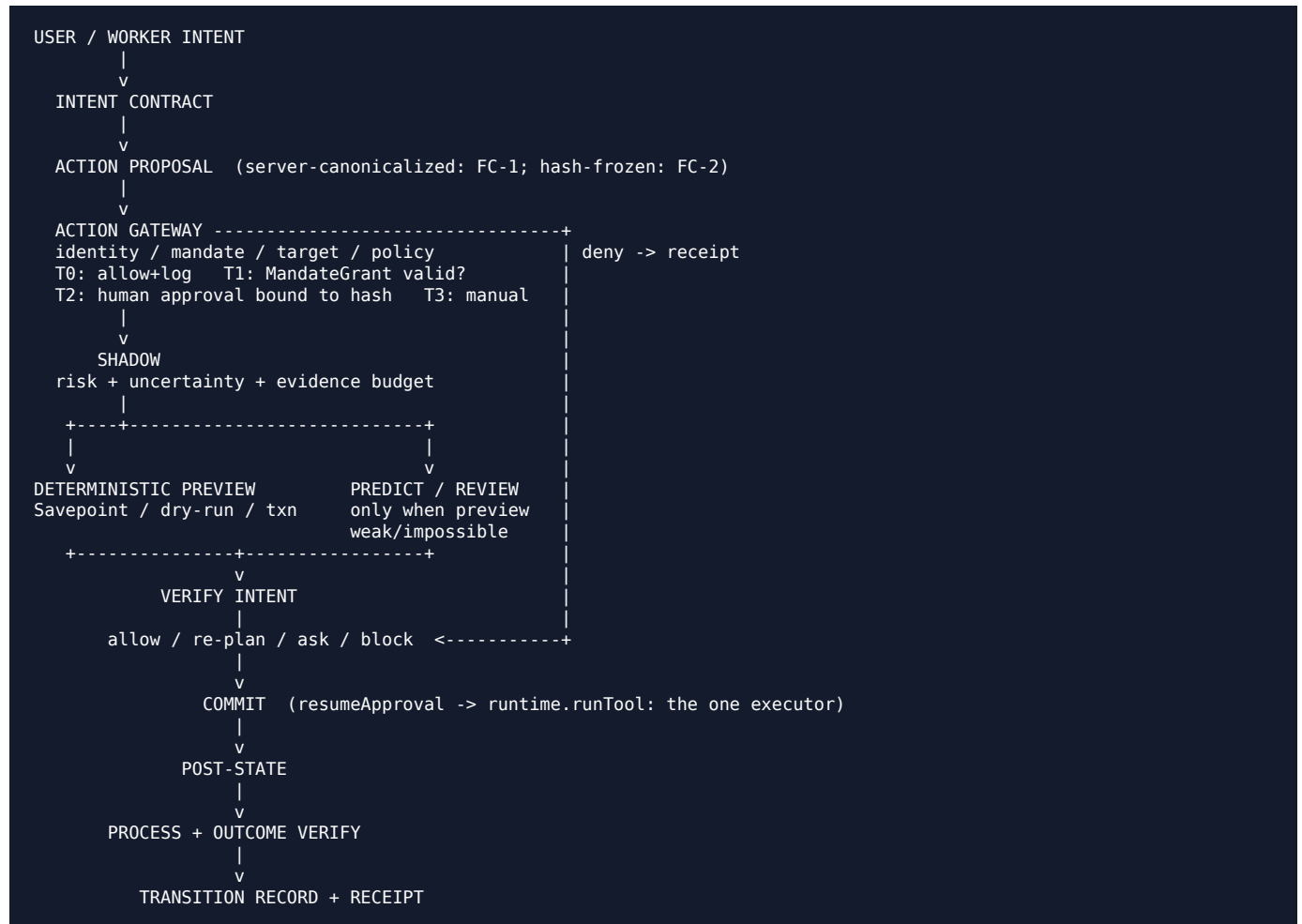
19. Spatial World Model Architecture



RUVIEW TRUST RULE

RuView remains observe-only until JARVIS reproduces useful real-world performance against trusted sensors. Repository implementation and synthetic benchmarks are architecture signals, not evidence that our room deployment is accurate. A RuView observation can never be the sole evidence for a T1 or higher action.

20. Action, Consequence & Verification Pipeline



21. Authority & Safety Model v5

Class	Examples	Default
T0 read / observe	status, read files, query room/world, summaries	Allowed when data scope permits; logged.
T1 bounded reversible	lights, local note, reversible workspace edits	Executes only under (a) frozen Phase 22 standing consent for voice, or (b) a valid MandateGrant v1 for a worker - plus policy and verification read. Every execution receipted.
T2 consequential reversible/external	send message, push branch, install dependency, external write, issue a mandate, promote a procedure	Worker/voice may propose; explicit UI confirmation with the human-session marker is always required. No exceptions clause.
T3 destructive/sensitive	delete valuable state, credentials, governance/trust changes, reclassifying an action class between tiers , financial	Manual only. Worker cannot initiate or self-authorize.
Physical guidance	R&D haptic/motor output	Separate safety class; deterministic hardware bounds and physical stop mandatory.

- Policy outside the model: prompts may request, never authorize (frozen; Phase 24 FC-1 is the mechanism).
- Deny before allow; missing/invalid policy fails closed.
- Approval binds an exact normalized action/plan hash and expires; it is not an open-ended permission token (frozen: FC-2, re-validated at decision time).
- **A mandate is a Gate artifact.** Minted only through the approval lifecycle as a T2; bound to actor, action class, target class, budget and expiry; revocable at any time with immediate effect; checked at execution time; receipted per execution. Workers, schedulers and nodes cannot mint, widen, renew or delegate one. Missing, expired or ambiguous mandate = deny.
- Worker delegation can only narrow authority, never expand it.
- **Tier reclassification is T3.** The set of actions in each tier is versioned configuration; moving an action class down a tier is a governance change with a registry entry and a closeout test.
- Credentials remain local to the node/runtime where possible; secrets are write-only and redacted from conversational and generic telemetry surfaces.
- Physical/room state actions require verification reads or post-state observation where hardware supports it.
- RuView and other experimental sensors cannot be sole evidence for T1+ actions.
- Cloud escalation requires explicit data-class/privacy policy and is always observable.
- Ingested text of any origin (proposal bodies, worker messages, MCP results, documents, transcripts) is DATA, never instructions (EoP-11, frozen).

22. Testing, Benchmarks & Closeout Gates

Test family	Required bar
Unit/integration/properly	100% pass for touched runtime and all frozen invariant suites (Phase 21 battery, Phase 23 closeout, Phase 24 non-bypass drill).
Adapter conformance	Fake and real adapters pass the same contract before the real path is enabled.
Disabled-feature guards	Forbidden authority/capture/bypass paths structurally fail closed; the disabled-feature matrix is re-asserted.
Mutation-path canary	runtime.runTool = 2 production call sites; resumeApproval sole executor; spine byte-frozen; MCP import allowlist holds (GATE-2).
E2E user journeys	Real UI/voice/mobile/browser/node flows pass on the primary machine.
Chaos/failure	Kill process, drop node/network, expire approval, revoke mandate mid-run, deny permission, corrupt provider response; recover deterministically.
Privacy canaries	Inject secrets/PII/raw frames and prove they do not escape into generic telemetry/UI.
Routing benchmark	Accuracy + latency + cost + reliability by task class on current hardware.
Behavioral stress	Repeated failure, deadline pressure, poisoned docs, contradictory instructions, tool outages.
Shadow autonomy benchmark	Damaging transition rate, completion, false block, approvals per success, latency/cost, recovery success.
Spatial benchmark	Sensor accuracy, disagreement, latency, world-state staleness, false gesture/action rate.
Voice benchmark	Wake false-positive/negative, end-to-end latency, interruption, consent enforcement.
Long-duration worker	Multi-hour task + process restart + node loss + checkpoint resume.
Soak	Phase 25H: seven consecutive days of real use from the packaged build, receipts as evidence.
Closeout hygiene	tsc --noEmit, lint at baseline, hooks in full, no --no-verify, closeout doc + registry flip + recap in the same commit, no co-author trailer.
Regression baseline	Every phase stores reproducible benchmark artifacts before closeout.

23. Superseded Decisions -> Registry Amendments

v4 listed these as replaced. In v5 each is a registry amendment: it names the frozen invariant it touches and the shape the change must take. The precedent is FC-2, which wrapped the frozen Phase 18 lifecycle additively rather than adding a lifecycle state. No slice opens against an old position until its amendment row exists.

Old position	v5 replacement	Frozen invariant touched	Required amendment shape
'No cloud agent that runs while laptop sleeps.'	Cloud continuity allowed as an explicitly governed runtime driver after durable task/placement controls exist.	Local-first; cloud opt-in, budgeted, logged.	Track E driver behind kill switch; per-task cloud policy; T2 blocks on phone approval; secrets never leave local.
'Suggestion-only agents, none execute.'	Preserved for unbounded/background proposals. Workers may perform T1 inside a MandateGrant; T2/T3 unchanged.	Approval lifecycle is the only path to side effects.	MandateGrant v1 minted as a T2; execution-time check; receipt per execution; chaos test revokes mid-run.
'Scheduler is suggestion-only.' (Phase 8)	Scheduler may wake a worker on a durable schedule.	Phase 8: no routine tool calls, execution, auto approvals, chaining.	Additive wrap: wake event != authorization; a test proves a woken worker with no mandate can only read and propose.
'One agent / no cross-agent chaining.'	Governed coordinator -> specialists -> reviewer with typed durable peer messages.	No authority expansion; ingested content is data.	WorkerMessage bodies are data; delegation only narrows; peer bus cannot reach the executor (import allowlist pattern).
'Memory improves through summaries/retrieval only.'	Verified experiences and procedures, human-promoted.	No silent writes; derived state is not canonical truth.	Promotion inbox (D8); PromotionDecision carries a human-session marker.
'Vision = individual CV tools.'	Canonical sensor observations + fused Spatial World Model.	I2-I4 (no raw frames, gated providers, no background capture).	27A contract freeze; privacy canaries per sensor.
'MacBook target = M4 Air 16GB.' / 'M3 Pro 18GB.'	M1 Max 32GB confirmed; 28A measures its envelope.	none	Registry records the machine at 25D and the measured envelope at 28A; no roadmap number is authoritative before that.
'UI read-only forever.'	Approvals, steering, take-wheel and governed controls become explicit user surfaces where the authority model requires them.	I5: pipeline/audit surfaces carry no execute/approve/mutate affordances.	Approval surfaces are a separate, AMBER-marked class rendering canonical effects; I5 stays true for evidence surfaces; no run/retry affordances anywhere.
'MCP is an integration feature.'	MCP is one transport behind the capability/action gateway; it never owns authority.	Phase 24 frozen server; consuming external MCP out of scope.	B2b with its own threat model; the 24 exposure matrix untouched.
'Router picks model tier only.'	Router also picks worker topology, evidence budget and execution node.	Capability-based routing; cost caps.	28B additive fields; explainable decisions.
'Success = tool/model returned successfully.'	Process success and outcome success permanently separated.	none (strengthens Phase 21A Verification Agent)	C5 verifier; Receipt carries both flags.

24. Instructions to Claude Code / Codex Before Any New Implementation

FIRST MOVE

Do not start Program U, Phase 25, or any Expansion Track from this PDF alone. Run R.1 first: a fresh repository reality audit against current main. The system sat for roughly eight weeks while architecture and external dependencies moved. The audit maps current code to sections 3, 6 and 7 and preserves what is already real instead of rebuilding it.

- **Read, in this order:** REGISTRY.md; the current recap; PHASE23_ENTRY_AUDIT.md and PHASE23_CLOSEOUT_VERIFICATION.md; MCP_GATEWAY_THREAT_MODEL.md and MCP_GATEWAY_ARCHITECTURE.md; UI_CAPSTONE_BRIEF.md; DESIGN.md and PRODUCT.md; EXPANSION_ERA_REFRESH.md and EXPANSION_ERA_V2.md; prompts/e011-voice-consolidation.md; then this document.
- **Numbering-collision check:** list every phase number and registry ID present in docs, tests and prompts. Confirm 22-24 are frozen with the scopes in section 7. Confirm nothing claims 25-30. Report CLEAR or the collision; a collision halts everything until the registry resolves it.
- **Count check:** runtime.runTool production call sites = 2; resumeApproval is the sole executor; spine byte-identical to the Phase 21 baseline; MCP import allowlist (GATE-2) green.
- **Hardware check:** confirm the M1 Max 32GB in the audit (chip, RAM, storage, macOS version) and list every Windows-path or CUDA assumption in scripts, sidecars and tests; do not carry the roadmap's envelope prior into any config.
- Classify every capability with Foundation Complete / Workflow Complete / Realized / Execution Enabled / Operationally Validated, mapped to registry states.
- Identify what can be directly reused for WorkerProfile/TaskEnvelope/NodeCapability/ActionProposal/MandateGrant/TransitionRecord before creating new stores or buses. Do not add another scheduler, event bus, policy engine or memory database until the audit proves the existing subsystem cannot satisfy the contract.
- Any external dependency (Hermes/OpenBot/OpenClaw/AG-UI/OpenShell/AgentCore/RuView/Soup) enters behind an adapter/driver with a kill switch and conformance tests, and its version is re-verified at adapter-entry.
- One active slice at a time. Single agent, no subagents, no Agent-tool fan-out for building. Parallel read-only probing is allowed only for testing and only with Prince's approval.
- No commits without standing authority, and then only when every verification is green; no --no-verify; no co-author trailers. Each slice ends with exact status, test counts, and a suggested commit message.
- Every prompt: role line first, YES/NO lists naming specific systems, observations mapped to tools, numbers on all spatial values, explicit DO-NOTs, explicit invariants, explicit tests, the registry entry it serves.

25. Recommended Immediate Execution Sequence

Order	Action	Reason
0	R.1 fresh repository reality audit (incl. numbering, count and hardware checks)	Prevents rebuilding completed capabilities; catches drift; settles the three confirmations.
1	R.2 stabilize: tests, dependencies, primary-machine bootstrap, flake family	Green baseline before architecture movement.
2	R.3 E-011 voice surface consolidation	The live next slice since July; the capstone renders voice state.
3	R.4 real-vs-synthetic check (light); R.5 camera proof when hardware present	The capstone expresses truth, not phantoms.
4	Program U (U.1-U.7) -> Execution Enabled	Phase 1-24 capabilities deserve a usable Command Center before new complexity.
5	Phase 25 Productionization (25A-25H); gate OPEN	The DONE phase. Real data, voice live, bootstrap, packaging, soak. Everything after it is measured on this baseline.
6	28A benchmark slice, measurement-only	Informs 26A voice and 28B routing; not a separate phase.
7	Phase 26 Presence & Voice	Highest daily-use value; fast-brain/deep-worker split.
8	Phase 27 Spatial World Model	One perception substrate before more sensors/gauntlet.
9	Phase 28 Local Intelligence/Council	Dynamic evidence routing and structured outputs.
10	Phase 29 Smart Room/Spatial Interaction	The world model now has something real to control.
11	Phase 30 Mobile/Node Fabric	Mac/Windows/iPhone coherent execution and supervision surfaces.
12	Track A Durable Worker Runtime	Persistence and lifecycle become real (absorbs live steering).
13	Track B Governed Computer (+B2b MCP client) + Track C Shadow	Durable workers get useful, constrained execution.
14	Track D Procedure Learning (human-promoted)	Only after runtime + Shadow can verify learned workflows.
15	Track E Cloud Continuity	Only after checkpoint/placement/authority are proven locally.
16	Track F Spatial Gauntlet	Built on validated world model and node fabric.
17	Track G Human augmentation research	Last, optional, safety-gated research only.

WHAT SHIPS NEXT (anchor)

R.1 audit prompt, then **R.2**, then **E-011**. Nothing in sections 10-16 is buildable until those three are COMPLETE in the registry. If a session ends with a new idea for a later phase, it goes into the registry as PROPOSED and nowhere else.

26. Milestone Demonstrations That Prove the Architecture

Demo	What must happen
Numbering audit proof (new)	R.1 report shows the phase/registry map with 22-24 frozen at their real scopes and 25-30 unclaimed. Nothing else opens without it.
Productionization soak (new)	Fresh clone on the primary machine -> bootstrap from README -> packaged build -> seven days of daily use with receipts -> chaos drills recover.
Presence demo	Wake phrase -> instant acknowledgement -> complex repo question -> conversation continues while a deep worker runs -> result interrupts naturally.
Mandate demo (new)	Human mints a bounded mandate (T2 UI confirm) -> worker executes T1 under it with receipts -> mandate revoked mid-run -> next execution denied with a receipt -> worker attempts to widen the mandate -> rejected, audited.
Worker persistence demo	Builder starts a migration -> API/runtime restarts -> Builder resumes from durable checkpoint with the same intent and mandate.
Node demo	Task starts on Mac -> Windows node performs an Unreal/Windows step -> phone shows progress and can steer/approve -> artifact returns to Mac.
Take-wheel demo	Browser worker reaches 2FA -> asks for help -> phone/desktop handoff -> user completes login -> worker resumes; secret absent from transcript.
Shadow demo	Worker proposes risky repo cleanup -> Savepoint previews -> Shadow detects hidden harm -> re-plans -> safe final state passes verifier -> receipt shows the prevented transition. The human approved the final hash; Shadow approved nothing.
Procedure demo	User demonstrates a workflow once -> candidate -> replay/verify -> promotion inbox -> human promotes -> later executes from one short instruction.
Spatial demo	Depth/RGB + second sensor update one room model -> hand manipulates a virtual object -> audio/haptic feedback follows -> room state reacts only after a governed action.
Continuity demo	Mac sleeps -> approved cloud-safe portion continues -> phone supervises -> local-only final step waits -> Mac returns -> job resumes and finishes.
Trust demo	Inject poisoned memory/source -> candidate quarantined -> canonical memory and procedure registry unchanged -> the poisoned candidate appears in the inbox labelled untrusted and cannot be promoted without an explicit override that is itself audited.
Hardware augmentation research demo	Only if ever approved: symbolic motor intent bounded by deterministic controller + watchdog + physical emergency stop; the model cannot emit raw stimulation parameters.

27. Source Basis & Research Signals

Internal JARVIS documents used as architecture truth: Master Architecture Roadmap v3/v3.1; ARCHITECTURE_OPERATIONALIZATION; EXPANSION_ERA, EXPANSION_ERA_REFRESH (03 Jun 2026), EXPANSION_ERA_V2; JARVIS UI Polish Plan and UI_CAPSTONE_BRIEF; PHASE23_ENTRY_AUDIT and PHASE23_CLOSEOUT_VERIFICATION; MCP_GATEWAY_THREAT_MODEL and MCP_GATEWAY_ARCHITECTURE (14 Jun 2026); the Enhancement Registry; the Phase 1-9 handoff; the July 2026 build sessions (Phase 24 freeze, WorkflowBox v1a-v1c, E-011 scoping, hardware envelope decision).

Internal research/project sources: Shadow / Fable Technical Strategy (Savepoint v0.1, Intent Contract, Transition Record, deterministic counterfactual execution, evidence budget); Sleepwalk & Lucid dissertation plan (provenance-aware consolidation, quarantine, promotion gates, behavioral checks); AffectiveSec research direction.

External references (status verified 01 Sep 2026; re-verify at adapter-entry)

Reference	URL	Verified status
OpenBot architecture / governed computers	github.com/CopilotKit/openbot (docs/architecture.md)	Real. First tagged release v0.0.1 on 17 Aug 2026; project self-describes as alpha. Pattern-adopt only.
OpenClaw repository, nodes, automations	github.com/openclaw/openclaw (docs/nodes, docs/automation/cron-jobs.md)	As cited in v4; re-verify at adapter-entry.
AG-UI protocol	github.com/ag-ui-protocol/ag-ui	As cited in v4; re-verify at adapter-entry.
Hermes Agent releases	github.com/NousResearch/hermes-agent/releases	v0.21.0 'Pantheon' released 31 Aug 2026 (bot mode, cron memory, subagent steering, MCP command center). High churn; pattern-adopt only.
NVIDIA OpenShell / NemoClaw	developer.nvidia.com blog: run autonomous self-evolving agents more safely with NVIDIA OpenShell	As cited in v4; re-verify at adapter-entry.
Amazon Bedrock AgentCore runtime instances	aws.amazon.com what's-new 2026/08; docs.aws.amazon.com bedrock-agentcore runtime instances	GA Aug 2026; managed EC2, sessions up to 14 days; microVM option up to 8 hours. Evaluate in Track E.
RuView unified RF spatial world model	github.com/ruvnet/RuView docs/adr/ADR-273 and ADR index	As cited in v4; observe-only regardless.
Human Operator research prototype	humanoperator.org; github.com/danielkajzer/Human-Operator	As cited in v4; Track G only.

28. Final Architectural Directive

THE SENTENCE TO KEEP

JARVIS should feel like one intelligence, but operate like a governed system: fast presence at the surface; persistent specialist Workers underneath; models and runtimes treated as replaceable resources; one world model for physical context; one action gateway for authority; Shadow/Savepoint for consequence verification; and a promotion firewall with a human at its door so experience compounds without silently corrupting the system.

- Do not rebuild Hermes, OpenBot, OpenClaw, AgentCore or RuView inside JARVIS. Borrow their strongest primitives behind our contracts, and take no runtime dependency on alpha software.
- Do not sacrifice the governance spine for convenience. External runtimes are replaceable; JARVIS identity, authority, memory policy, world model and evidence semantics are ours.
- Do not call scaffold complete. A feature that has never run in reality on the primary machine, on main, with a closeout doc, is not done.
- Do not let the model be its own policy engine, scheduler, credential store, verifier, promoter, mandate-issuer or physical safety controller.
- Do not over-engineer speculative future layers ahead of the next operational proof. Phase 25 is that proof for the whole system; each later layer earns itself through a real demo.

- Do not reuse a phase number history already owns. Numbers are cheap; reopening a frozen phase by accident is not.
- Use the primary Mac + the Windows laptop as a heterogeneous local cluster first. Cloud comes later as an execution driver, not as the brain that owns JARVIS.

JARVIS v5: LOCAL-FIRST. PERSISTENT. SPATIAL. CONSEQUENCE-AWARE. VERIFIED ON MAIN.